AI Engineer — Take-Home Assignment

PhD Shortlist Builder

Time: 72 hours from receipt **Submission:** GitHub link via Google Form (link in covering email)

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1. Context

Ambitio helps international students apply to PhD programs abroad. A core part of our product is the **PhD Shortlist** — a personalised list of supervisors, programs, and funded opportunities surfaced for each student, used downstream to draft personalised cold-emails to professors.

For each student, we need to produce a shortlist of **~50–200 actionable supervisor + program matches** that a domain mentor would unhesitatingly approve as worth contacting.

The challenge is not the LLM call. The challenge is the **data**: surfacing the right humans, with the right evidence, in the right context, with no embarrassing mismatches.

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2. Goal

Build a system that ingests a **student profile** and produces a **ranked shortlist of PhD supervisors + programs**, each with grant/paper evidence and a personalised `why_match` that a student can reference when emailing the professor.

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3. Input

A student profile JSON containing:

- Education history (degrees, institutions, grades, theses)
- Skills, projects, publications
- Research interests (3–5 stated areas)
- Target countries (hard constraint)
- Target intake (semester + year)
- Intro call summary (free text)
- Raw resume text

4. Output

For each student, produce a single JSON shortlist:

- 50–200 supervisor recommendations
- For each: name, institution, country, contact email (if obtainable), research focus, evidence (papers and/or grants), `why_match` blurb
- Tier (reach / target / safety) if relevant
- Linked PhD programs or open positions
- A consistent, documented schema

5. Requirements

#	Requirement
1	**Coverage** — ≥50 actionable recommendations spread across the student's stated areas
2	**Country adherence** — 100% within target countries (hard constraint)
3	**Evidence** — every supervisor must carry verifiable paper(s) or grant(s) with links
4	**Personalisation** — `why_match` must reference specific PI work that maps onto the student's profile, not generic praise
5	**Machine-readable output** — consistent JSON schema, documented in the README
6	**Reproducibility** — given the same input, the system should run end-to-end with a single command
7	**Latency** — target wall-clock < 15 minutes per shortlist on a single laptop / one cloud VM

6. Data Quality Challenges to Consider

The following are **real failure modes** we have observed. Your design should address as many as you can in 72 hours. We do not expect you to solve all of them — we will grade on which ones you noticed, what trade-offs you chose, and how well you justified them.

6.1 Same-name-different-person collisions

"Yang Shi", "Yu Meng", "Ying Ma", "Wei Wang", "Rong Zheng", "Sharma" are extremely common. The supervisor you surface may be confused with an unrelated researcher of the same name. Catching the mistake from a paper title alone is not enough — you need to verify the human matches the student's research area before trusting them.

6.2 Career-stage errors

PhD students and fresh postdocs appear in author databases with their own first-author papers, but they cannot supervise PhDs. If you treat any name in an author list as a "PI", you will surface 24-year-old grad students. Personal-award fellowships (NIH F31/F32, UKRI studentships, MSCA postdoc grants) list the **awardee** — usually a junior researcher, not the supervisor.

6.3 Wrong-domain leakage from keyword overlap

- A grant titled "biodegradable plastic cartridges" can leak into a **biomaterials** student's list — it is actually military ammunition R&D.
- A grant on "high-elevation social-ecological systems" can leak into a **Himalayan pilgrimage** student's list — it is actually Pacific Northwest fire archaeology.
- A "trauma-informed" grant can leak into a **clinical psychology** student's list — it is actually a literary-history project on grief in Roman antiquity.
- A "DNA barcoding" grant can leak into a **plant biology** student's list — it is actually single-cell barcoding for human Hi-C chromatin work.

How do you tell **discipline** (humanities vs STEM vs medical) and **region** (which country / continent / ecosystem) from a grant abstract before deciding it matches a student?

6.4 Eligibility filters in free-text ads

Many PhD vacancies have citizenship/residency restrictions ("UK only", "home fees", "EU residents") buried in the ad. Surfacing an ineligible position to an Indian student is worse than not surfacing it. Consider how you would extract eligibility from unstructured ad text.

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7. How We Grade

We will run your system against 1-2 real student profiles and evaluate:

Dimension	Method
Mentor-eye audit	A domain mentor reviews your top 30 picks and rates each as bullseye / solid / stretch / wrong. We compute (bullseye + solid) / 30. Past systems score 60–85%.
Contamination count	Number of clear wrong-domain / wrong-person / non-PI entries out of total. Past systems: 5–20%.
Coverage	≥50 recommendations spread across stated areas.
Country adherence	100% in target countries. Hard fail if breached.
Latency	Wall-clock per shortlist.

Dimension	Method
Process quality	Your `DECISIONS.md` — what trade-offs you saw, what you chose, why. We weight this heavily for a 72-hour exercise.

We weight contamination heavier than coverage. Surfacing a wrong-person is worse than missing a borderline-correct one. An 80%-mentor-approved shortlist of 60 PIs beats a 60%-mentor-approved shortlist of 150 PIs.

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8. Deliverables

A **GitHub repository** (public, or shared privately with the email in the covering message) containing:

1. `**README.md` ** — your approach overview, data sources, how to run, design trade-offs, known limitations.
2. **Code** that runs end-to-end on a provided sample student profile with one command.
3. `**sample_output/<student_id>.json` ** — at least one full shortlist JSON for a sample student profile we send you.
4. `**DECISIONS.md` **— 1–2 page writeup explaining how you addressed** at least 5** of the 15 data quality challenges above, with concrete examples from your own output. We read this carefully.
5. `**schema.md` ** — your output JSON schema, documented.

Submit the GitHub repository URL via the Google Form linked in your covering email.

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9. Constraints & Permissions

- You may use **any** commercial API, open dataset, scraping, LLM, vector DB, or framework you want.
- **Cite your data sources** clearly in the README.
- Use of LLMs is expected; we are evaluating system design, not whether you can call an API.
- You may NOT submit code/output produced by another candidate or shared between candidates.
- If you spend more than 3 days, you are over-investing — we are grading the trade-offs you made in a constrained window.

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10. Bonus — Closing the Feedback Loop

Imagine your shortlist has been used. Each student emailed some subset of the supervisors you surfaced. Weeks or months later, we receive back a CSV with outcomes for every email sent:

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student_id, supervisor_id, institution, area, sent_at, outcome
106419,    A5031856973,    UNSW,         PTSD,  2026-07-12, ADMIT
106419,    A5055000569,    UNSW,         PTSD,  2026-07-12, REJECT
106419,    A5089892421,    Ohio State,   Pilgrim, 2026-07-12, NO_REPLY
...
```

Possible `outcome` values include: `ADMIT`, `INTERVIEW`, `POSITIVE_REPLY`, `REJECT`, `NO_REPLY`, `BOUNCE`, `OUT_OF_OFFICE`, `WRONG_PERSON`, `NOT_RECRUITING`.

Design question

How would you build a system that uses this outcome stream to **automatically improve future shortlists**?

We are not looking for a single "right" answer. We want to see how you reason about it. Consider — and pick the angles you find most interesting:

Deliverable for this bonus

A **working implementation** that ingests the outcomes CSV and improves the system for next shortlist and documentation around approach

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